TRIUNE BIBLICAL UNIVERSITY GLOBAL EXTENSION USA

Bachelor of Artificial Intelligence and Robotics Engineering

# AI.304

## Robot Operating Systems

### COURSE OVERVIEW

The ROS ecosystem: nodes, topics, services, actions, launch files, and middleware for robot software.

ROBOT OPERATING SYSTEMS

1

SESSION

 60 MINUTES

# ORIENTATION: THE ROBOT OS

**KEY TOPICS**

- What is ROS
- ROS architecture
- Why ROS dominates robotics
- ROS1 vs ROS2

**TEACHING FOCUS**

Learn the middleware of modern robots.

▲ **KEY TAKEAWAY** ROS is the shared language of robot software.

ROBOT OPERATING SYSTEMS

2

SESSION

⏱ 60 MINUTES

## NODES & COMMUNICATION

**KEY TOPICS**

- ROS nodes
- Topics and publishers/subscribers
- Services
- Messages and parameters

**TEACHING FOCUS**

Build modular robot software.

▲ **KEY TAKEAWAY** Nodes talk through topics and services.

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3

SESSION

60 MINUTES

## TOOLS & WORKFLOW

**KEY TOPICS**

- roscore, rosrun, roslaunch
- rviz and gazebo
- rosbag
- Catkin/colcon build

**TEACHING FOCUS**

Develop, visualize, and log robot software.

**KEY TAKEAWAY** Tools make ROS development tractable.

4

SESSION

60 MINUTES

# ACTIONS & COORDINATION

**KEY TOPICS**

- Actionlib and action servers
- Goal feedback and preemption
- Complex robot behaviors
- State machines (SMACH)

**TEACHING FOCUS**

Manage long-running robot tasks.

**KEY TAKEAWAY** Actions coordinate multi-step behavior.

5

SESSION

60 MINUTES

## ROS IN PRACTICE

**KEY TOPICS**

- Integrating drivers
- Multi-robot namespaces
- Real-time considerations
- Debugging ROS

**TEACHING FOCUS**

Deploy ROS on real robots.

**KEY TAKEAWAY** ROS must run reliably on hardware.

ROBOT OPERATING SYSTEMS

6

SESSION

⏱ 60 MINUTES

# ROS PROJECT & REVIEW

**KEY TOPICS**

- ROS-based robot demo
- Course recap and assessment
- Toward cognitive robotics (AI.305)
- Code review

**TEACHING FOCUS**

Build a complete ROS robot system.

▲ **KEY TAKEAWAY** ROS integrates perception, planning, and action.